

# Measure Theory Axler Notes

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## 1 Section 2A

### 1.1 Basics/Definitions

**Definition** (Length of Open Interval;  $\ell(I)$ ). The **length**  $\ell(I)$  of an open interval  $I$  is defined by

$$\ell(I) = \begin{cases} b - a & \text{if } I = (a, b) \text{ for some } a, b \in \mathbb{R} \text{ with } a < b \\ 0 & \text{if } I = \emptyset \\ \infty & \text{if } I = (-\infty, a) \text{ or } I = (a, \infty) \text{ for some } a \in \mathbb{R} \\ \infty & \text{if } I = (-\infty, \infty) \end{cases}$$

**Definition** (Outer Measure;  $|A|$ ). The **outer measure**  $|A|$  of a set  $A \subseteq \mathbb{R}$  is defined by

$$|A| = \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) : I_1, I_2, \dots \text{ are open intervals such that } A \subset \bigcup_{k=1}^{\infty} I_k \right\}.$$

## 1.2 Good Properties of Outer Measure

**Proposition** (Countable Sets Have Outer Measure 0). Every countable subset of  $\mathbb{R}$  has outer measure 0.

**Proposition** (Outer Measure Preserves Order). Suppose  $A$  and  $B$  are subsets of  $\mathbb{R}$  with  $A \subset B$ . Then  $|A| \leq |B|$ .

**Proof.** ■

**Definition** (Translation;  $t + A$ ). If  $t \in \mathbb{R}$  and  $A \subset \mathbb{R}$ , then the **translation**  $t + A$  is defined by

$$t + A = \{t + a : a \in A\}.$$

**Proposition** (Outer Measure is Translation Invariant). Suppose  $t \in \mathbb{R}$  and  $A \subset \mathbb{R}$ . Then  $|t + A| = |A|$ .

**Proposition** (Countable Subadditivity of Outer Measure). Suppose  $A_1, A_2, \dots$  is a sequence of subsets of  $\mathbb{R}$ . Then

$$\left| \bigcup_{k=1}^{\infty} A_k \right| \leq \sum_{k=1}^{\infty} |A_k|.$$

**Proof.** ■

**Definition** (Open Cover). Suppose  $A \subseteq \mathbb{R}$ .

- A collection  $\{O_\alpha\}_{\alpha \in \Lambda}$  of open subsets of  $\mathbb{R}$  is called an **open cover** of  $A$  if  $A$  is contained in the union of all the sets in  $\{O_\alpha\}_{\alpha \in \Lambda}$ .
- An open  $\{O_\alpha\}_{\alpha \in \Lambda}$  of  $A$  is said to have a **finite subcover** if  $A$  is contained in the union of some finite list of sets in  $\{O_\alpha\}_{\alpha \in \Lambda}$ .

**Proposition** (Heine-Borel Theorem). Every open cover of a closed bounded subset of  $\mathbb{R}$  has a finite subcover.

## 1.3 Outer Measure of Closed Bounded Interval

**Proposition** (Outer Measure of a Closed Interval). Suppose  $a, b \in \mathbb{R}$ , with  $a < b$ . Then  $|[a, b]| = b - a$ .

**Proposition** (Nontrivial Intervals are Uncountable). Every interval in  $\mathbb{R}$  that contains at least two distinct elements is uncountable.

## 1.4 Outer Measure is Not Additive

**Proposition** (Nonadditivity of Outer Measure). There exist disjoint subsets  $A$  and  $B$  of  $\mathbb{R}$  such that

$$|A \cup B| \neq |A| + |B|.$$

## 2 Section 2B

### 2.1 Nonexistence of Extension of Length to All Subsets of $\mathbb{R}$

**Proposition** (Nonexistence of Extension of Length to All Subsets of  $\mathbb{R}$ ). There does not exist a function  $\mu$  with all the following properties:

- (a)  $\mu$  is a function from the set of subsets of  $\mathbb{R}$  to  $[0, \infty]$ ,
- (b)  $\mu(I) = \ell(I)$  for every open interval  $I$  of  $\mathbb{R}$ ,
- (c)  $\mu\left(\bigcup_{k=1}^n A_k\right) = \sum_{k=1}^n \mu(A_k)$  for every disjoint sequence  $A_1, A_2, \dots$  of subsets of  $\mathbb{R}$ ,
- (d)  $\mu(t + A) = \mu(A)$  for every  $A \subseteq \mathbb{R}$  and every  $t \in \mathbb{R}$ .

### 2.2 $\sigma$ -Algebra

**Definition** ( $\sigma$ -Algebra). Suppose  $X$  is a set and  $\mathcal{S}$  is a set of subsets of  $X$ . Then  $\mathcal{S}$  is called a  $\sigma$ -algebra on  $X$  if the following three conditions are satisfied:

- $\emptyset \in \mathcal{S}$ ;
- if  $E \in \mathcal{S}$ , then  $X \setminus E \in \mathcal{S}$ ;
- if  $E_1, E_2, \dots$  is a sequence of elements of  $\mathcal{S}$ , then  $\bigcup_{k=1}^{\infty} E_k \in \mathcal{S}$ .

**Proposition** ( $\sigma$ -algebras are Closed Under Countable Intersection). Suppose  $\mathcal{S}$  is a  $\sigma$ -algebra on a set  $X$ . Then

- (a)  $X \in \mathcal{S}$ ;
- (b) if  $D, E \in \mathcal{S}$ , then  $D \cup E \in \mathcal{S}$  and  $D \cap E \in \mathcal{S}$  and  $D \setminus E \in \mathcal{S}$ ;
- (c) if  $E_1, E_2, \dots$  is a sequence of elements of  $\mathcal{S}$ , then  $\bigcap_{k=1}^{\infty} E_k \in \mathcal{S}$ .

**Definition** (Measurable Space; Measurable Set). • A **measurable space** is an ordered pair  $(X, \mathcal{S})$ , where  $X$  is a set and  $\mathcal{S}$  is a  $\sigma$ -algebra on  $X$ .

- An element of  $\mathcal{S}$  is called an  **$\mathcal{S}$ -measurable set**, or just a **measurable set** if  $\mathcal{S}$  is clear from the context.

### 2.3 Borel Subsets of $\mathbb{R}$

**Proposition** (Smallest  $\sigma$ -algebra containing a collection of subsets). Suppose  $X$  is a set and  $\mathcal{A}$  is a set of subsets of  $X$ . Then the intersection of all  $\sigma$ -algebras on  $X$  that contain  $\mathcal{A}$  is a  $\sigma$ -algebra on  $X$ .

**Definition** (Borel Set). The smallest  $\sigma$ -algebra on  $\mathbb{R}$  containing all open subsets of  $\mathbb{R}$  is called the collection of **Borel subsets of  $\mathbb{R}$** . An element of this  $\sigma$ -algebra is called a **Borel set**.

**Definition** (Inverse Image;  $f^{-1}(A)$ ). If  $f : X \rightarrow Y$  is a function and  $A \subset Y$ , then the set  $f^{-1}(A)$  is defined by

$$f^{-1}(A) = \{x \in X : f(x) \in A\}.$$

**Proposition** (Algebra of Inverse Images). Suppose  $f : X \rightarrow Y$  is a function. Then

- (a)  $f^{-1}(Y \setminus A) = X \setminus f^{-1}(A)$  for every  $A \subset Y$ ;
- (b)  $f^{-1}(\bigcup_{A \in \mathcal{A}} A) = \bigcup_{A \in \mathcal{A}} f^{-1}(A)$  for every  $\mathcal{A}$  of subsets of  $Y$ ;
- (c)  $f^{-1}(\bigcap_{A \in \mathcal{A}} A) = \bigcap_{A \in \mathcal{A}} f^{-1}(A)$  for every  $\mathcal{A}$  of subsets of  $Y$ .

**Proposition** (Inverse Image of a Composition). Suppose  $f : X \rightarrow Y$  and  $g : Y \rightarrow W$  are functions. Then

$$(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A)) \quad \forall A \subset W.$$

## 2.4 Measurable Functions

**Proposition** (Condition for Measurable Function). Suppose  $(X, \mathcal{S})$  is a measurable space and  $f : X \rightarrow \mathbb{R}$  is a function such that

$$f^{-1}((a, \infty)) \in \mathcal{S} \quad \forall a \in \mathbb{R}.$$

Then  $f$  is an  $\mathcal{S}$ -measurable function.

**Definition** (Borel Measurable Function). Suppose  $X \subset \mathbb{R}$ . A function  $f : X \rightarrow \mathbb{R}$  is called **Borel measurable** if  $f^{-1}(B)$  is a Borel set for every  $B \subset \mathbb{R}$ .

**Proposition** (Every Continuous Function is Borel Measurable). Every continuous real-valued function defined on a Borel subset of  $\mathbb{R}$  is a Borel measurable function.

**Definition** (Increasing Function). Suppose  $X \subset \mathbb{R}$  and  $f : X \rightarrow \mathbb{R}$  is a function.

- $f$  is called **increasing** if  $f(x) \leq f(y)$  for all  $x, y \in X$  with  $x < y$ .
- $f$  is called **strictly increasing** if  $f(x) < f(y)$  for all  $x, y \in X$  with  $x < y$ .

**Proposition** (Every Increasing Function is Borel Measurable). Every increasing function defined on a Borel subset of  $\mathbb{R}$  is a Borel measurable function.

**Proposition** (Composition of Measurable Functions). Suppose  $(X, \mathcal{S})$  is a measurable space and  $f : X \rightarrow \mathbb{R}$  is an  $\mathcal{S}$ -measurable function. Suppose  $g$  is a real-valued Borel measurable function defined on a subset of  $\mathbb{R}$  that includes the range of  $f$ . Then  $g \circ f : X \rightarrow \mathbb{R}$  is an  $\mathcal{S}$ -measurable function.

**Proposition** (Algebraic Operations with Measurable Functions). Suppose  $(X, \mathcal{S})$  is a measurable space and  $f, g : X \rightarrow \mathbb{R}$  are  $\mathcal{S}$ -measurable. Then

- (a)  $f + g, f - g$ , and  $fg$  are  $\mathcal{S}$ -measurable functions;
- (b) if  $g(x) \neq 0$  for all  $x \in X$ , then  $\frac{f}{g}$  is an  $\mathcal{S}$ -measurable function.

**Proposition** (Limit of  $\mathcal{S}$ -measurable Functions). Suppose  $(X, \mathcal{S})$  is a measurable space and  $f_1, f_2, \dots$  is a sequence of  $\mathcal{S}$ -measurable functions from  $X$  to  $\mathbb{R}$ . Suppose  $\lim_{k \rightarrow \infty} f_k(x)$  exists for each  $x \in X$ . Define  $f : X \rightarrow \mathbb{R}$  by

$$f(x) = \lim_{k \rightarrow \infty} f_k(x).$$

Then  $f$  is an  $\mathcal{S}$ -measurable function.

**Definition** (Borel Subsets). A subset of  $[-\infty, \infty]$  is called a **Borel set** if its intersection with  $\mathbb{R}$  is a Borel set.

**Definition** (Measurable Function). Suppose  $(X, \mathcal{S})$  is a measurable space. A function  $f : X \rightarrow [-\infty, \infty]$  is called  $\mathcal{S}$ -measurable if  $f^{-1}(B) \in \mathcal{S}$  for every Borel set  $B \subset [-\infty, \infty]$ .

**Proposition** (Condition for Measurable Function). Suppose  $(X, \mathcal{S})$  is a measurable space and  $f : X \rightarrow [-\infty, \infty]$  is a function such that

$$f^{-1}((a, \infty]) \in \mathcal{S} \quad \forall a \in \mathbb{R}.$$

Then  $f$  is an  $\mathcal{S}$ -measurable function.

**Proposition** (Infimum and Supremum of a Sequence of  $\mathcal{S}$ -measurable Functions). Suppose  $(X, \mathcal{S})$  is a measurable space and  $f_1, f_2, \dots$  is a sequence of  $\mathcal{S}$ -measurable functions from  $X$  to  $[-\infty, \infty]$ . Define  $g, h : X \rightarrow [-\infty, \infty]$  by

$$g(x) = \inf_{k \in \mathbb{Z}^+} f_k(x) \text{ and } h(x) = \sup_{k \in \mathbb{Z}^+} f_k(x).$$

Then  $g$  and  $h$  are  $\mathcal{S}$ -measurable functions.

## 3 Section 2C

### 3.1 Definition of Measures

**Definition** (Measure). Suppose  $X$  is a set and  $\mathcal{S}$  is a  $\sigma$ -algebra on  $X$ . A **measure** on  $(X, \mathcal{S})$  is a function  $\mu : \mathcal{S} \rightarrow [0, \infty]$  such that  $\mu(\emptyset) = 0$  and

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k)$$

for every disjoint sequence  $E_1, E_2, \dots$  of sets in  $\mathcal{S}$ .

**Definition** (Measure Space). A **Measure Space** is an ordered triple  $(X, \mathcal{S}, \mu)$ , where  $X$  is a set,  $\mathcal{S}$  is a  $\sigma$ -algebra on  $X$ , and  $\mu$  is a measure on  $(X, \mathcal{S})$ .

### 3.2 Properties of Measures

**Proposition** (Measure Preserves Order; Measure of a Set Difference). Suppose  $(X, \mathcal{S}, \mu)$  is a measure space and  $D, E \in \mathcal{S}$  are such that  $D \subset E$ . Then

- (a)  $\mu(D) \leq \mu(E)$ ;
- (b)  $\mu(E \setminus D) = \mu(E) - \mu(D)$  provided that  $\mu(D) < \infty$ .

**Proposition** (Countable Subadditivity). Suppose  $(X, \mathcal{S}, \mu)$  is a measure space and  $E_1, E_2, \dots \in \mathcal{S}$ . Then

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) \leq \sum_{k=1}^{\infty} \mu(E_k).$$

**Proposition** (Measure of an Increasing Union). Suppose  $(X, \mathcal{S}, \mu)$  is a measure space and  $E_1 \subset E_2 \subset \dots$  is an increasing sequence of sets in  $\mathcal{S}$ . Then

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \lim_{k \rightarrow \infty} \mu(E_k).$$

**Proposition** (Measure of a Decreasing Intersection). Suppose  $(X, \mathcal{S}, \mu)$  is a measure space and  $E_1 \supset E_2 \supset \dots$  is a decreasing sequence of sets in  $\mathcal{S}$ , with  $\mu(E_1) < \infty$ . Then

$$\mu\left(\bigcap_{k=1}^{\infty} E_k\right) = \lim_{k \rightarrow \infty} \mu(E_k).$$

**Proposition** (Measure of Union). Suppose  $(X, \mathcal{S}, \mu)$  is a measure space and  $D, E \in \mathcal{S}$  with  $\mu(D \cap E) < \infty$ . Then

$$\mu(D \cup E) = \mu(D) + \mu(E) - \mu(D \cap E).$$

## 4 Section 2D

### 4.1 Additivity of Outer Measure on Borel Sets

**Proposition** (Additivity of Outer Measure if One of the Sets is Open). Suppose  $A$  and  $G$  are disjoint subsets of  $\mathbb{R}$  and  $G$  is open. Then

$$|A \cup G| = |A| + |G|.$$

**Proposition** (Additivity of Outer Measure if One of the Sets is Closed). Suppose  $A$  and  $F$  are disjoint

subsets of  $\mathbb{R}$  and  $F$  is closed. Then

$$|A \cup F| = |A| + |F|.$$

**Proposition** (Approximation of Borel Sets from Below by Closed Sets). Suppose  $B \subset \mathbb{R}$  is a Borel set. Then for every  $\varepsilon > 0$ , there exists a closed set  $F \subset B$  such that  $|B \setminus F| < \varepsilon$ .

**Proposition** (Additivity of Outer Measure if One of the Sets is a Borel Set). Suppose  $A$  and  $B$  are disjoint subsets of  $\mathbb{R}$  and  $B$  is a Borel set. Then

$$|A \cup B| = |A| + |B|.$$

**Proposition** (Existence of a subset of  $\mathbb{R}$  is not a Borel set). There exists a set  $B \subset \mathbb{R}$  such that  $|B| < \infty$  and  $B$  is not a Borel set.

**Proposition** (Outer Measure is a Measure on Borel Sets). Outer measure is a measure on  $(\mathbb{R}, \mathcal{B})$ , where  $\mathcal{B}$  is the  $\sigma$ -algebra on Borel subsets of  $\mathbb{R}$ .

**Definition** (Lebesgue Measure). **Lebesgue Measure** is the measure on  $(\mathbb{R}, \mathcal{B})$ , where  $\mathcal{B}$  is the  $\sigma$ -algebra of Borel subsets of  $\mathbb{R}$ , that assigns to each Borel set its outer measure.

## 4.2 Lebesgue Measurable Sets

**Definition** (Lebesgue Measurable Set). A set  $A \subset \mathbb{R}$  is called **Lebesgue Measurable** if there exists a Borel set  $B \subset A$  such that  $|A \setminus B| = 0$ .

**Proposition** (Equivalence for being a Lebesgue measurable set). Suppose  $A \subset \mathbb{R}$ . Then the following are equivalent:

- (a)  $A$  is Lebesgue measurable.
- (b) For each  $\varepsilon > 0$ , there exists a closed set  $F \subset A$  with  $|A \setminus F| < \varepsilon$ .
- (c) There exist closed sets  $F_1, F_2, \dots$  contained in  $A$  such that

$$\left| A \setminus \bigcup_{k=1}^{\infty} F_k \right| = 0.$$

- (d) There exists a Borel set  $B \subset A$  such that  $|A \setminus B| = 0$ .
- (e) For each  $\varepsilon > 0$ , there exists an open set  $G \supset A$  such that  $|G \setminus A| < \varepsilon$ .
- (f) There exist open sets  $G_1, G_2, \dots$  containing  $A$  such that

$$\left| \left( \bigcup_{k=1}^{\infty} G_k \right) \setminus A \right| = 0.$$

- (g) There exists a Borel set  $B \supset A$  such that  $|B \setminus A| = 0$ .

**Proposition** (Outer Measure is a measure on Lebesgue Measurable sets). (a) The set  $\mathcal{L}$  of Lebesgue measurable subsets of  $\mathbb{R}$  is a  $\sigma$ -algebra on  $\mathbb{R}$ .

(b) Outer measure is a measure on  $(\mathbb{R}, \mathcal{L})$ .

**Definition** (Lebesgue Measure). **Lebesgue Measure** is the measure on  $(\mathbb{R}, \mathcal{L})$ , where  $\mathcal{L}$  is the  $\sigma$ -algebra of Lebesgue measurable subsets of  $\mathbb{R}$ , that assigns to each Lebesgue measurable set its outer measure.

## 5 Section 2E

### 5.1 Pointwise and Uniform Convergence

**Definition** (Pointwise Convergence; Uniform Convergence). Suppose  $X$  is a set,  $f_1, f_2, \dots$  is a sequence of functions from  $X$  to  $\mathbb{R}$ , and  $f$  is a function from  $X$  to  $\mathbb{R}$ .

- The sequence  $f_1, f_2, \dots$  converges pointwise on  $X$  to  $f$  if

$$\lim_{k \rightarrow \infty} f_k(x) = f(x)$$

for each  $x \in X$ . That is, for each  $x \in X$  and every  $\varepsilon > 0$ , there exists an  $n \in \mathbb{Z}^+$  such that  $|f_k(x) - f(x)| < \varepsilon$  for all  $k \geq n$ .

- The sequence  $f_1, f_2, \dots$  **converges uniformly** on  $X$  to  $f$  if for every  $\varepsilon > 0$ , there exists an  $n \in \mathbb{Z}^+$  such that

$$|f_k(x) - f(x)| < \varepsilon \quad \forall k \geq n \text{ and } \forall x \in X.$$

**Proposition** (Uniform Limit of Continuous Functions is Continuous). Suppose  $B \subset \mathbb{R}$  and  $f_1, f_2, \dots$  is a sequence of functions from  $B$  to  $\mathbb{R}$  that converges uniformly on  $B$  to a function  $f : B \rightarrow \mathbb{R}$ . Suppose  $b \in B$  and  $f_k$  is continuous at  $b$  for each  $k \in \mathbb{Z}^+$ . Then  $f$  is continuous at  $b$ .

### 5.2 Egorov's Theorem

**Proposition** (Egorov's Theorem). Suppose  $(X, \mathcal{S}, \mu)$  is a measure space with  $\mu(X) < \infty$ . Suppose  $f_1, f_2, \dots$  is a sequence of  $\mathcal{S}$ -measurable functions from  $X \rightarrow \mathbb{R}$  that converges pointwise on  $X$  to a function  $f : X \rightarrow \mathbb{R}$ . Then for every  $\varepsilon > 0$ , there exists a set  $E \in \mathcal{S}$  such that  $\mu(X \setminus E) < \varepsilon$  and  $f_1, f_2, \dots$  converges uniformly to  $f$  on  $E$ .

### 5.3 Approximation by Simple Functions

**Definition** (Simple Function). A function is called **simple** if it takes on only finitely many values.

**Proposition** (Approximation by Simple Functions). Suppose  $(X, \mathcal{S})$  is a measure space and  $f : X \rightarrow [-\infty, \infty]$  is  $\mathcal{S}$ -measurable. Then there exists a sequence  $f_1, f_2, \dots$  of functions from  $X$  to  $\mathbb{R}$  such that

- (a) each  $f_k$  is a simple  $\mathcal{S}$ -measurable function;
- (b)  $|f_k(x)| \leq |f_{k+1}(x)| \leq |f(x)|$  for all  $k \in \mathbb{Z}^+$  and all  $x \in X$ ;



- (c)  $\lim_{k \rightarrow \infty} f_k(x) = f(x)$  for every  $x \in X$ ;
- (d)  $f_1, f_2, \dots$  converges uniformly on  $X$  to  $f$  if  $f$  is bounded.

## 5.4 Luzin's Theorem

**Proposition** (Luzin's Theorem). Suppose  $g : \mathbb{R} \rightarrow \mathbb{R}$  is a Borel measurable function. Then for every  $\varepsilon > 0$ , there exists a closed set  $F \subseteq \mathbb{R}$  such that  $|\mathbb{R} \setminus F| < \varepsilon$  and  $g|_F$  is a continuous function on  $F$ .

**Proposition** (Continuous Extensions of Continuous Functions). • Every continuous function on a closed subset of  $\mathbb{R}$  can be extended to a continuous function on all of  $\mathbb{R}$ .

- More precisely, if  $F \subset \mathbb{R}$  is closed and  $g : F \rightarrow \mathbb{R}$  is continuous, then there exists a continuous function  $h : \mathbb{R} \rightarrow \mathbb{R}$  such that  $h|_F = g$ .

**Proposition** (Luzin's theorem; Second Version). Suppose  $E \subseteq \mathbb{R}$  and  $g : E \rightarrow \mathbb{R}$  is a Borel measurable function. Then for every  $\varepsilon > 0$ , there exists a closed set  $F \subseteq E$  and continuous function  $h : \mathbb{R} \rightarrow \mathbb{R}$  such that  $|E \setminus F| < \varepsilon$  and  $h|_F = g|_F$ .

## 5.5 Lebesgue Measurable Functions

**Definition** (Lebesgue Measurable Functions). A function  $f : A \rightarrow \mathbb{R}$ , where  $A \subseteq \mathbb{R}$ , is called **Lebesgue Measurable** if  $f^{-1}(B)$  is a Lebesgue measurable set for every Borel set  $B \subseteq \mathbb{R}$ .

**Proposition** (Every Lebesgue Measurable Function is Almost Borel Measurable). Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is a Lebesgue measurable function. Then there exists a Borel measurable function  $g : \mathbb{R} \rightarrow \mathbb{R}$  such that

$$|\{x \in \mathbb{R} : g(x) \neq f(x)\}| = 0.$$

## 6 Section 4A

### 6.1 Markov's Inequality

**Proposition** (Markov's Inequality). Suppose  $(X, \mathcal{S}, \mu)$  is a measure space and  $h \in \mathcal{L}^1(\mu)$ . Then

$$\mu(\{x \in X : |h(x)| \geq c\}) \leq \frac{1}{c} \|h\|_1$$

for every  $c > 0$ .

**Definition** (3 Times A Bounded Open Interval). Suppose  $I$  is a bounded nonempty open interval of  $\mathbb{R}$ . Then  $3 * I$  denotes the open interval with the same center as  $I$  and three times the length of  $I$ .

**Lemma** (Vitali Covering Lemma). Suppose  $I_1, \dots, I_n$  is a list of bounded nonempty open intervals of  $\mathbb{R}$ . Then there exists a disjoint sublist  $I_{k_1}, \dots, I_{k_m}$  such that

$$I_1 \cup \dots \cup I_n \subseteq (3 * I_{k_1}) \cup \dots \cup (3 * I_{k_m}).$$

## 6.2 Hardy-Littlewood Maximal Inequality

**Definition** (Hardy-Littlewood Maximal Function). Suppose  $h : \mathbb{R} \rightarrow \mathbb{R}$  is a Lebesgue measurable function. Then the Hardy-Littlewood maximal function of  $h$  is the function  $h^* : \mathbb{R} \rightarrow [0, \infty]$  defined by

$$h^*(b) = \sup_{t>0} \frac{1}{2t} \int_{b-t}^{b+t} |h|.$$

**Proposition** (Hardy-Littlewood Inequality). Suppose  $h \in \mathcal{L}^1(\mathbb{R})$ . Then

$$|\{b \in \mathbb{R} : h^*(b) > c\}| \leq \frac{3}{c} \|h\|_1$$

for every  $c > 0$ .

## 7 Section 4B

**Theorem** (Lebesgue Differentiation Theorem, First Version). Suppose  $f \in \mathcal{L}^1(\mathbb{R})$ . Then

$$\lim_{t \rightarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} |f - f(b)| = 0$$

for almost every  $b \in \mathbb{R}$ .

**Definition** (Derivative). Suppose  $g : I \rightarrow \mathbb{R}$  is a function defined on an open interval  $I$  of  $\mathbb{R}$  and  $b \in I$ . The **derivative** of  $g$  at  $b$ , denoted  $g'(b)$ , is defined by

$$g'(b) = \lim_{t \rightarrow 0} \frac{g(b+t) - g(b)}{t}$$

if the limit above exists, in which case  $g$  is called **differentiable** at  $b$ .

**Theorem** (Fundamental Theorem of Calculus). Suppose  $f \in \mathcal{L}^1(\mathbb{R})$ . Define  $g : \mathbb{R} \rightarrow \mathbb{R}$  by

$$g(x) = \int_{-\infty}^x f.$$

Suppose  $b \in \mathbb{R}$  and  $f$  is continuous at  $b$ . Then  $g$  is differentiable at  $b$  and  $g'(b) = f(b)$ .

**Theorem** (Lebesgue Differentiation Theorem, second version). Suppose  $f \in \mathcal{L}^1(\mathbb{R})$ . Define  $g : \mathbb{R} \rightarrow \mathbb{R}$  by

$$g(x) = \int_{-\infty}^x f.$$

Then  $g'(b) = f(b)$  for almost every  $b \in \mathbb{R}$ .

**Proposition** (No Set Constitutes Exactly Half of Each Interval). There does not exist a Lebesgue measurable set  $E \subseteq [0, 1]$  such that

$$|E \cap [0, b]| = \frac{b}{2}$$

for all  $b \in [0, 1]$ .

**Proposition** ( $\mathcal{L}^1$  function equals its local average almost everywhere). Suppose  $f \in \mathcal{L}^1(\mathbb{R})$ . Then

$$f(b) = \lim_{t \rightarrow 0} \frac{1}{2t} \int_{b-t}^{b+t} f$$

for almost every  $b \in \mathbb{R}$ .